Week 2: Fundamental and Option Alphas

- Date: 2026-10-17
- Scope: BRAIN Tutorial Chapters 3–4
- Simulations: 19, 2 submitted
- Common settings: USA · D1 · Subindustry · Decay 4 · Truncation 0.08 (noted where different)

Summary

This week I moved beyond price–volume data. Chapter 3 used **financial-statement data** (assets, operating income, EBITDA); Chapter 4 used **option data** (implied volatility). The biggest lesson: *how* you compare a number matters as much as *which* number you use. The same earnings yield went from Sharpe 0.79 to 1.54 when I compared each company with its own past instead of with other companies. I submitted two alphas: a fundamental one in the TOP200 universe and an option + fundamental blend in TOP3000.

Roadmap

```
Chapter 3 – Fundamentals
  rank(equity/assets)      Sharpe -1.93  --flip sign-->  +1.93
  rank(op_income/cap)      Sharpe  0.79  (cross-sectional)
  ts_rank(op_income/cap)   Sharpe  1.54  (time-series)  --> works in TOP1000/500/200
                                                        --> SUBMITTED (TOP200, Sharpe 1.64)
Chapter 4 – Options
  call IV - put IV         Sharpe  1.91  but turnover 44%, fitness 0.81
  + group neutralization   Sharpe  2.15  fails: weight concentration, sub-universe
  + blend with fundamentals + NaN-safe add
                           Sharpe  2.49  fitness 1.69  --> SUBMITTED
```

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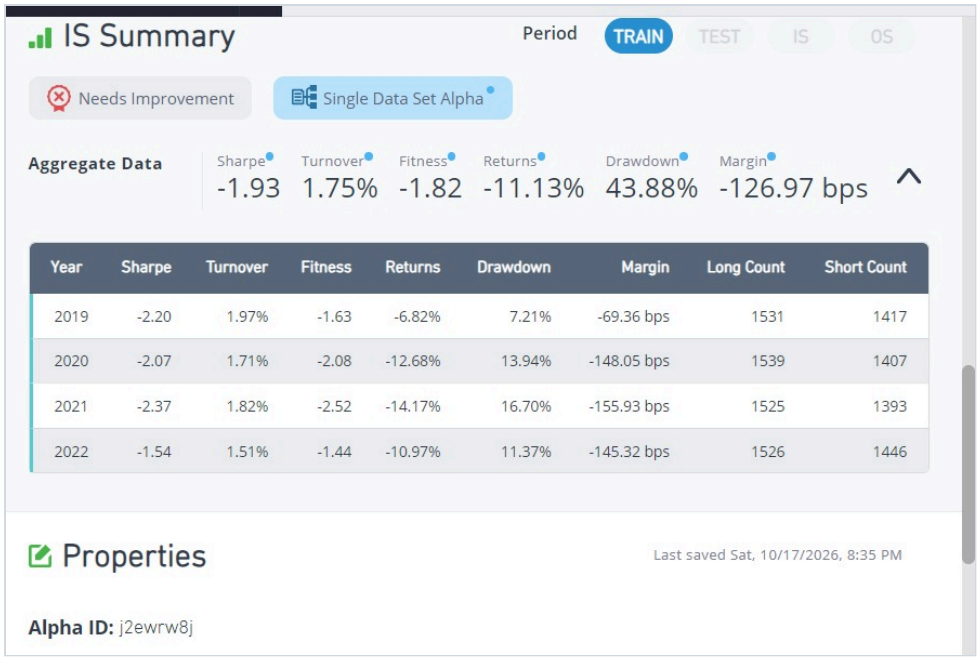
1. First fundamental alpha: leverage — Sharpe −1.93 → +1.93

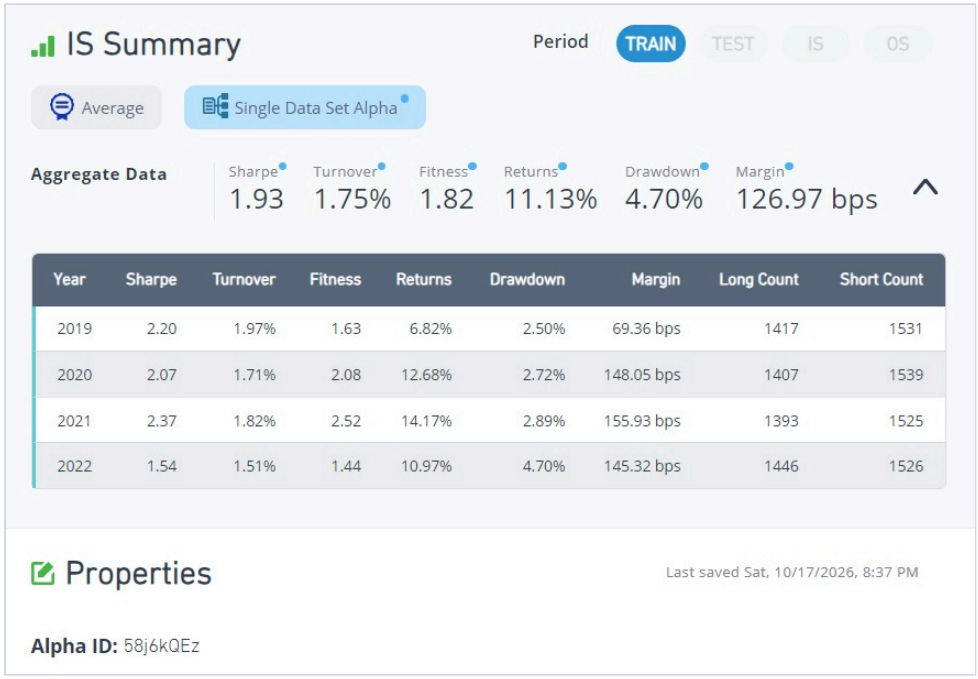
Idea: companies with a high equity-to-assets ratio (little debt) are healthier and should outperform.

Expression	Sharpe	Turnover	Fitness	Drawdown
`rank(equity/assets)`	-1.93	1.75%	-1.82	43.88%
`rank(-equity/assets)`	**1.93**	1.75%	**1.82**	**4.70%**

The hypothesis was wrong, and consistently so: negative in every year from 2019 to 2022. Flipping the sign gave a clean alpha. Within the same subindustry, **more leveraged companies did better** in this period, probably because cheap debt amplified returns in a low-rate, rising market.

Turnover was under 2%. Financial statements update quarterly, so fundamental signals barely trade. Price–volume alphas from week 1 traded 30–130%.



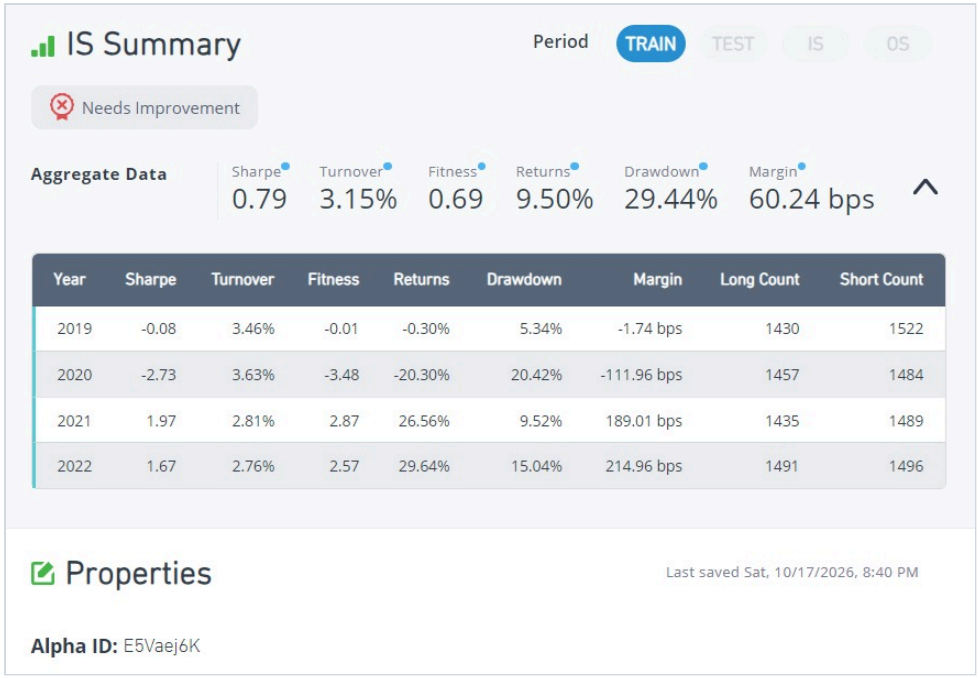


2. Value signals and why they swing with the market

Expression	Idea	Sharpe	2019	2020	2021	2022
`rank(operating_income/cap)`	Earnings yield: profit relative to market value	0.79	-0.08	**-2.73**	1.97	1.67
`rank(ebitda/enterprise_value)`	Inverse of EV/EBITDA (debt- and cash-adjusted)	0.45	-1.13	-1.53	1.45	1.56

Both are classic "buy cheap companies" signals, and both broke down in 2019–2020, when growth stocks led the market and cheap stocks crashed during COVID. They recovered in 2021–2022 when value came back. Ranking companies *against each other* by cheapness means always holding the same group of value stocks, so the alpha rides the value-vs-growth cycle.

`ebitda` is missing for about a third of the universe (banks, for example), so the second alpha traded only ~2,000 stocks.



IS Summary

Period TRAIN TEST IS OS

Needs Improvement

Aggregate Data

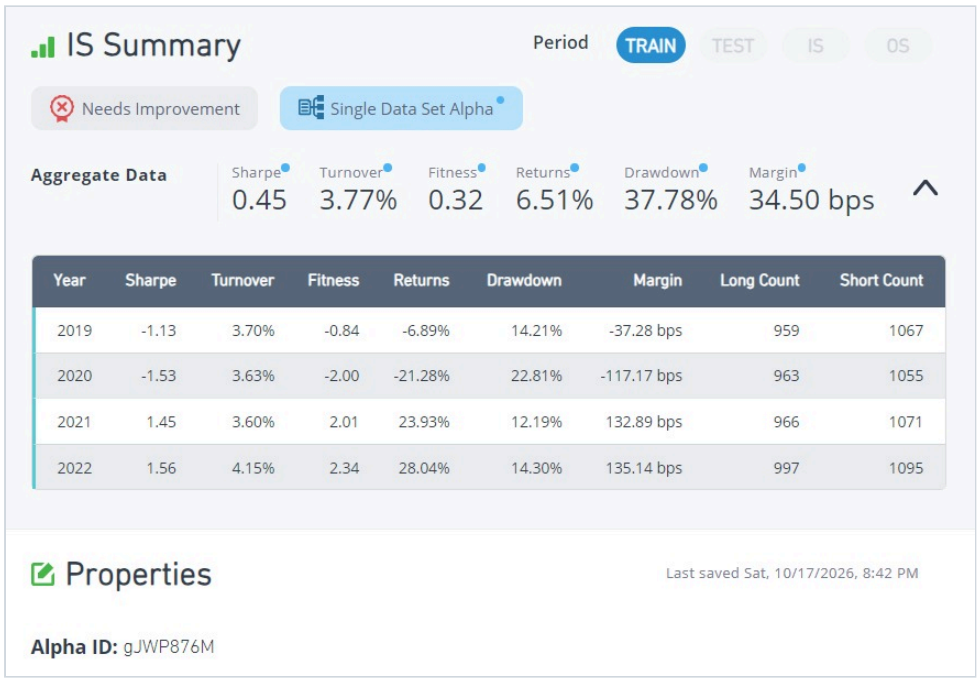
Sharpe	Turnover	Fitness	Returns	Drawdown	Margin
0.79	3.15%	0.69	9.50%	29.44%	60.24 bps

Year	Sharpe	Turnover	Fitness	Returns	Drawdown	Margin	Long Count	Short Count
2019	-0.08	3.46%	-0.01	-0.30%	5.34%	-1.74 bps	1430	1522
2020	-2.73	3.63%	-3.48	-20.30%	20.42%	-111.96 bps	1457	1484
2021	1.97	2.81%	2.87	26.56%	9.52%	189.01 bps	1435	1489
2022	1.67	2.76%	2.57	29.64%	15.04%	214.96 bps	1491	1496

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IS Summary

Period TRAIN TEST IS OS

Needs Improvement Single Data Set Alpha

Aggregate Data

Sharpe	Turnover	Fitness	Returns	Drawdown	Margin
0.45	3.77%	0.32	6.51%	37.78%	34.50 bps

Year	Sharpe	Turnover	Fitness	Returns	Drawdown	Margin	Long Count	Short Count
2019	-1.13	3.70%	-0.84	-6.89%	14.21%	-37.28 bps	959	1067
2020	-1.53	3.63%	-2.00	-21.28%	22.81%	-117.17 bps	963	1055
2021	1.45	3.60%	2.01	23.93%	12.19%	132.89 bps	966	1071
2022	1.56	4.15%	2.34	28.04%	14.30%	135.14 bps	997	1095

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3. Time-series vs cross-sectional: Sharpe 0.79 → 1.54

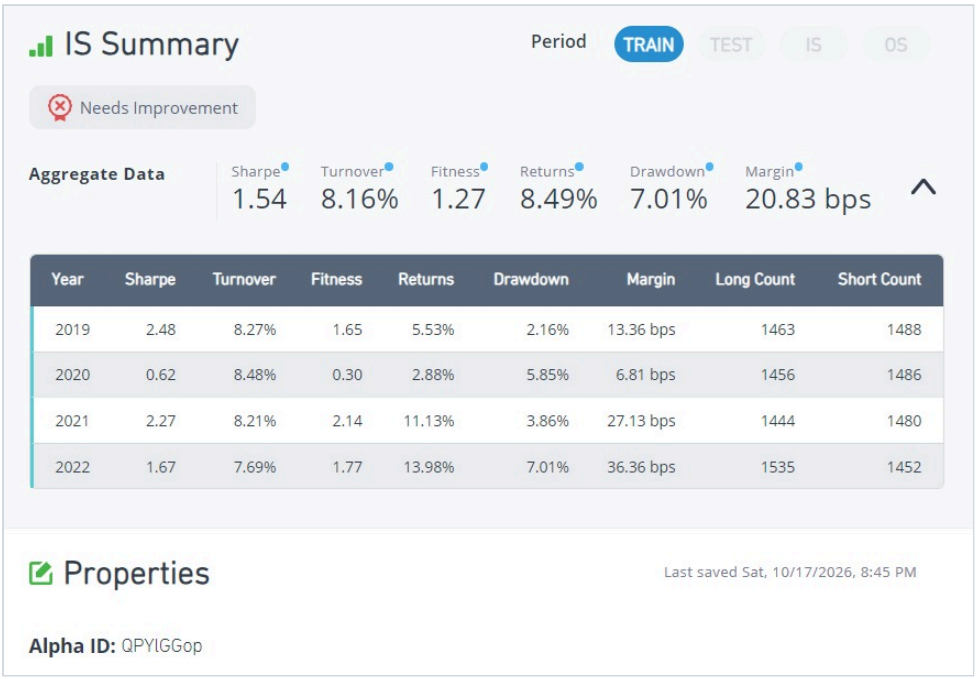
The tutorial's analogy: a student who scores 80 can be compared with their own past scores (time-series) or with classmates' scores (cross-sectional).

Type	Operators	Question it asks
Cross-sectional	`rank`, `zscore`, `winsorize`	Is this company cheap *compared with other companies today*?
Time-series	`ts_rank`, `ts_zscore`, `ts_mean`	Is this company cheap *compared with its own past*?

I applied `ts_rank` to the same operating earnings yield (OEY = operating_income / cap):

Expression	Sharpe	Turnover	Fitness	Drawdown	2020 Sharpe
`rank(operating_income/cap)`	0.79	3.15%	0.69	29.44%	-2.73
`ts_rank(operating_income/cap, 250)`	**1.54**	8.16%	**1.27**	**7.01%**	**+0.62**
`ts_rank(operating_income/cap, 126)`	1.51	10.28%	1.13	6.05%	+1.11

Comparing each company with its own last year removes the "always long value stocks" tilt, so the 2020 crash disappeared. A one-year window kept turnover lower than a six-month one at the same Sharpe.



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4. Does it survive in large caps? Universe test

Smaller universes keep only large, liquid stocks, where information is priced in faster, so signals usually get weaker. BRAIN's **sub-universe test** checks exactly this.

Universe	Sharpe	Turnover	Fitness	Returns	Stocks held
TOP3000	1.54	8.16%	1.27	8.49%	~2,950
TOP1000	1.75	8.79%	1.59	10.26%	~940
TOP500	1.60	8.98%	1.50	10.99%	~440
TOP200	**1.79**	9.43%	**2.07**	16.78%	~160

The opposite of what I expected: the signal held up, and was even stronger, in the largest 200 stocks. Large caps also react when their earnings yield improves against their own history.

IS Summary

Period: TRAIN | TEST | IS | OS

Average

Aggregate Data

Sharpe	Turnover	Fitness	Returns	Drawdown	Margin
1.75	8.79%	1.59	10.26%	4.69%	23.33 bps

Year	Sharpe	Turnover	Fitness	Returns	Drawdown	Margin	Long Count	Short Count
2019	2.05	8.95%	1.50	6.70%	1.78%	14.97 bps	461	478
2020	1.65	9.16%	1.46	9.77%	4.01%	21.35 bps	457	488
2021	1.50	8.69%	1.22	8.23%	4.69%	18.95 bps	472	465
2022	2.06	8.41%	2.35	16.31%	3.92%	38.77 bps	498	451

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IS Summary

Period: TRAIN | TEST | IS | OS

Needs Improvement

Aggregate Data

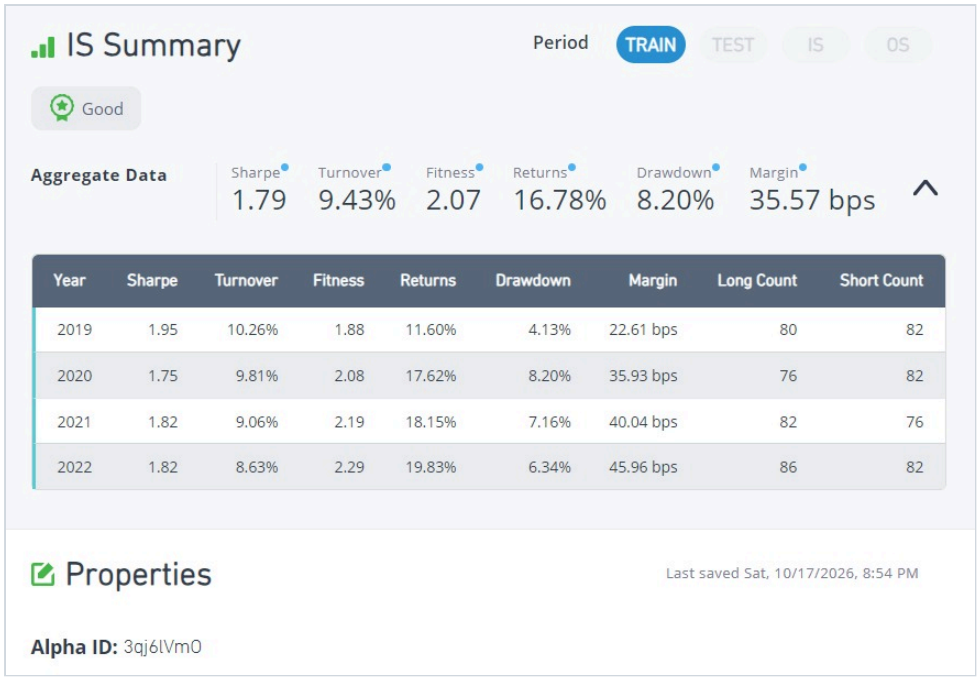
Sharpe	Turnover	Fitness	Returns	Drawdown	Margin
1.60	8.98%	1.50	10.99%	6.74%	24.48 bps

Year	Sharpe	Turnover	Fitness	Returns	Drawdown	Margin	Long Count	Short Count
2019	1.03	9.32%	0.61	4.41%	4.35%	9.47 bps	213	223
2020	2.64	9.37%	3.13	17.58%	2.64%	37.53 bps	209	225
2021	1.14	8.58%	0.87	7.35%	6.74%	17.14 bps	223	215
2022	1.66	8.70%	1.83	15.19%	6.15%	34.94 bps	230	215

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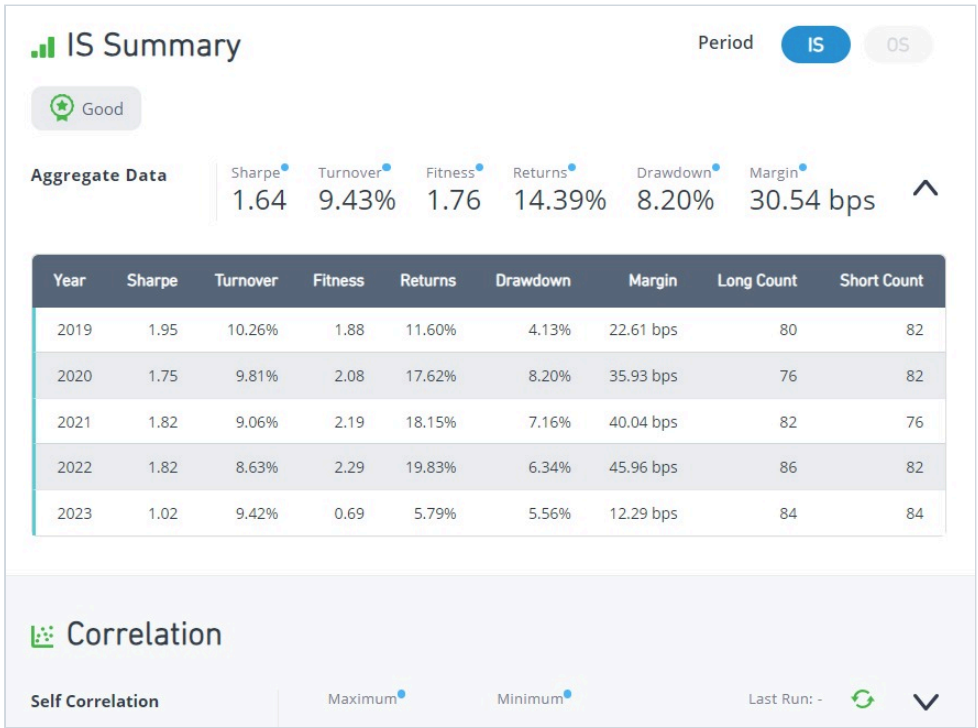
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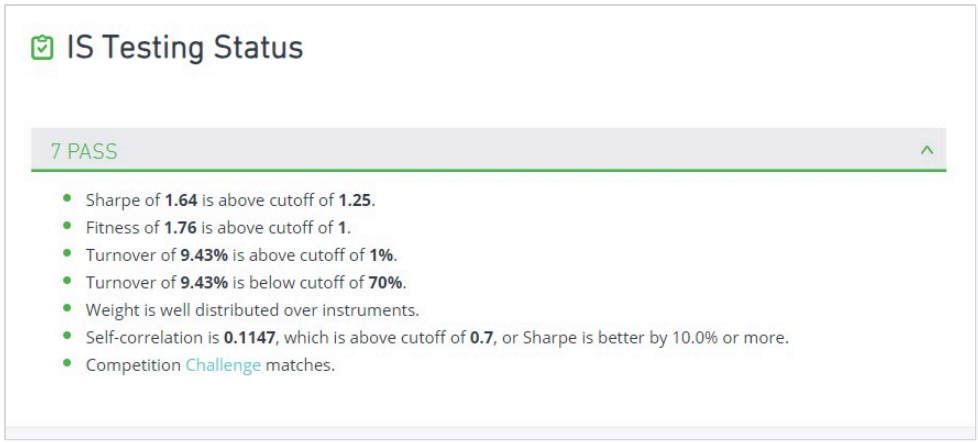


Submitted: OEY time-series rank in TOP200 ✓

Item	Detail
Expression	The tutorial's OEY `ts_rank` alpha above (250-day window)
Settings	**TOP200** · Subindustry · Decay 4 · Test period 0
Result (2019–2023)	**Sharpe 1.64 · Fitness 1.76 · Turnover 9.43%** · Returns 14.39% · Drawdown 8.20%
Checks	7 passed, self-correlation with my week 1 alpha 0.11

Setting test period to 0 shows the full in-sample period that the submission check actually uses (a week 1 lesson). 2023 was weaker (Sharpe 1.02), pulling the full-period Sharpe from 1.79 to 1.64, but still well above the 1.25 cutoff.







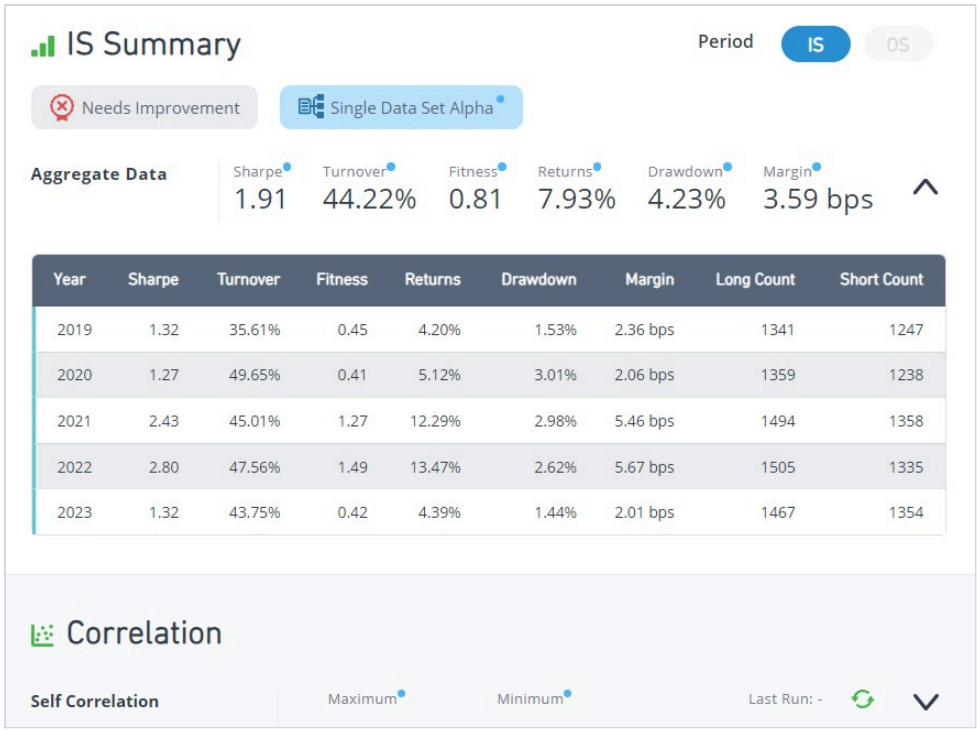
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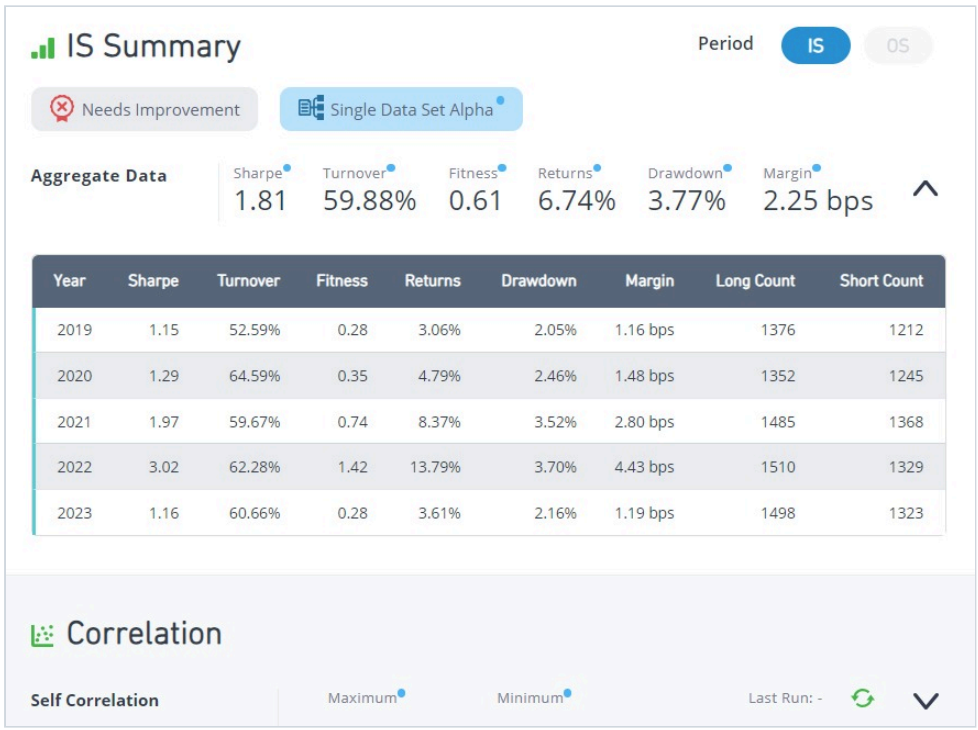
5. Option alphas: what option traders expect

Idea: option traders are usually better informed than stock traders. Implied volatility (IV) reflects option prices, so if call options are priced richer than puts (call IV > put IV), traders expect the stock to rise.

Expression	Maturity	Sharpe	Turnover	Fitness
`rank(implied_volatility_call_120 - implied_volatility_put_120)`	120 days	**1.91**	44.22%	0.81
same with 30-day options (assigned to a variable `iv_difference`)	30 days	1.81	59.88%	0.61

The signal was strong and positive every year, but IV changes daily, so turnover was in price–volume territory and fitness fell below 1. Short-dated IV is noisier and only added trading.





6. Group neutralization with custom buckets

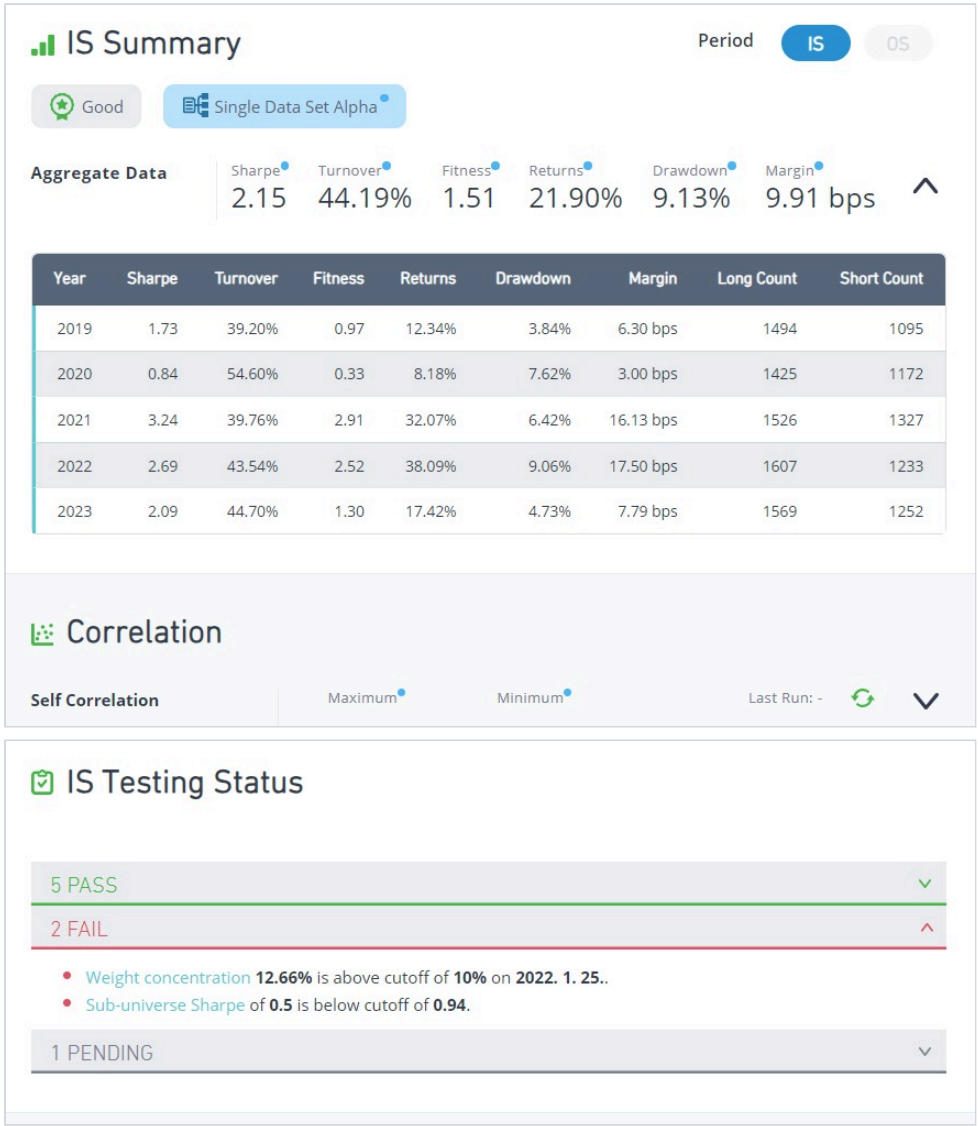
Small and volatile stocks naturally have larger IV gaps. To compare like with like, I grouped stocks into deciles with `bucket` and neutralized within each group.

```
iv_difference = implied_volatility_call_120 - implied_volatility_put_120;
vol_group     = bucket(rank(historical_volatility_120), range="0.1,1,0.1");
group_neutralize(iv_difference, vol_group)
```

Grouping	Sharpe	Fitness	Returns
None (section 5)	1.91	0.81	7.93%
Market-cap deciles	2.18	1.50	20.92%
Historical-volatility deciles	2.15	1.51	21.90%

Neutralization nearly doubled fitness. But it still **failed two submission checks**:

- **Weight concentration 12.66%** (limit 10%) on a single day, 2022-01-25
- **Sub-universe Sharpe 0.5** (needs 0.94): the signal lives mostly in small caps



7. Getting the option alpha to pass — Sharpe 2.49 ✓

Try	Change	Sharpe	Fitness	Failed checks	What I learned
1	Wrap the IV gap in `rank`	2.05	0.86	3	Concentration was **not** caused by outliers. Ranking erased useful magnitude
2	`ts_backfill(·, 20)` to fill gaps	1.30	0.92	3	Option data has long gaps in 2022–23; backfilling held stale values (2023 Sharpe −0.51)
3	Blend 50:50 with the OEY time-series signal	2.12	1.16	1	The fundamental signal, strong in large caps, **fixed the sub-universe test** and cut turnover to 31%
4	Turn on NaN handling	2.12	1.16	1	Identical result; the setting did not help
5	Replace `+` with `add(x, y, filter=true)`	**2.49**	**1.69**	**0**	`+` returns NaN if either side is NaN, so on days with no option data almost every stock dropped out. `filter=true` treats NaN as 0

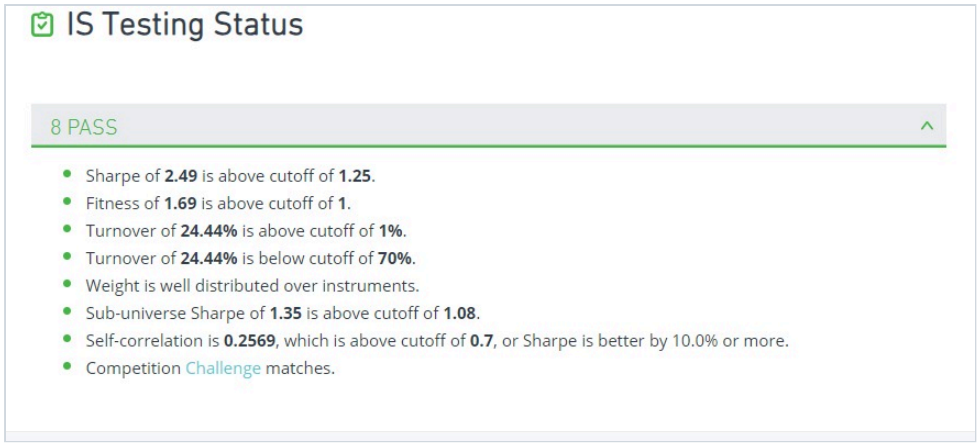
Submitted: option + fundamental blend ✓

Item	Detail
Idea	50% call–put IV gap (neutralized within volatility deciles) + 50% OEY time-series rank, combined NaN-safely
Operators	`rank`, `bucket`, `group_neutralize`, `ts_rank`, `add`
Data fields	`implied_volatility_call_120`, `implied_volatility_put_120`, `historical_volatility_120`, `operating_income`, `cap`
Settings	TOP3000 · Subindustry · Decay 4 · NaN handling On · Test period 0
Result (2019–2023)	**Sharpe 2.49 · Fitness 1.69 · Turnover 24.44%** · Returns 11.20% · Drawdown 4.41%
Checks	8 passed. Sub-universe Sharpe 1.35, self-correlation 0.26

> Since this alpha is my own submitted combination, I don't publish the exact expression.



Year	Sharpe	Turnover	Fitness	Returns	Drawdown	Margin	Long Count	Short Count
2019	3.03	20.33%	2.04	9.22%	1.43%	9.07 bps	1516	1488
2020	1.23	26.04%	0.63	6.87%	2.97%	5.28 bps	1500	1493
2021	4.31	26.06%	3.61	18.29%	1.79%	14.04 bps	1524	1508
2022	3.59	25.58%	3.12	19.27%	2.64%	15.06 bps	1573	1489
2023	0.59	24.33%	0.18	2.26%	4.41%	1.86 bps	1553	1525





Both submitted alphas flattened in 2023 (Sharpe 1.02 and 0.59). Both use the earnings-yield time-series signal, which may have weakened that year.

What I learned this week

1. **Fundamental data barely trades.** Turnover was 2–10%, compared with 30–130% for price–volume alphas.
2. **Compare with the company's own past.** `ts_rank` beat `rank` on the same earnings yield (Sharpe 0.79 → 1.54) and removed the value-cycle crash.
3. **A wrong hypothesis can still be a good alpha.** If a signal is consistently negative, flip it, then ask why.
4. **Read the failed check before tuning.** The concentration failure came from missing data on one day, not from outliers, so the fix was `add(..., filter=true)`, not `rank`.
5. **Blending signals with different strengths works again.** As in week 1, the fundamental leg covered the option signal's weak spot (large caps).

Next week

- Explore analyst and sentiment data fields
- Try universes other than TOP3000 for lower correlation between my alphas
- Investigate why both submitted alphas weakened in 2023

All experiments

#	Expression	Universe	Sharpe	Turnover	Fitness	Note
1	`rank(equity/assets)`	TOP3000	-1.93	1.75%	-1.82	
2	`rank(-equity/assets)`	TOP3000	1.93	1.75%	1.82	
3	`rank(operating_income/cap)`	TOP3000	0.79	3.15%	0.69	
4	`rank(ebitda/enterprise_value)`	TOP3000	0.45	3.77%	0.32	
5	`ts_rank(operating_income/cap, 250)`	TOP3000	1.54	8.16%	1.27	
6	`ts_rank(operating_income/cap, 126)`	TOP3000	1.51	10.28%	1.13	
7	#5	TOP1000	1.75	8.79%	1.59	
8	#5	TOP500	1.60	8.98%	1.50	
9	#5	TOP200	1.79	9.43%	2.07	
10	#5, test period 0	TOP200	1.64	9.43%	1.76	Submitted
11	`rank(iv_call_120 - iv_put_120)`	TOP3000	1.91	44.22%	0.81	Test period 0 from here
12	same, 30-day maturity	TOP3000	1.81	59.88%	0.61	
13	IV gap, market-cap bucket neutralized	TOP3000	2.18	44.36%	1.50	2 checks failed
14	IV gap, volatility bucket neutralized	TOP3000	2.15	44.19%	1.51	2 checks failed
15	#14 with `rank`	TOP3000	2.05	44.78%	0.86	3 checks failed
16	#14 with `ts_backfill`	TOP3000	1.30	27.43%	0.92	3 checks failed
17	#14 blended with OEY `ts_rank`	TOP3000	2.12	30.72%	1.16	1 check failed
18	#17, NaN handling On	TOP3000	2.12	30.72%	1.16	1 check failed
19	#18 with `add(filter=true)`	TOP3000	2.49	24.44%	1.69	Submitted